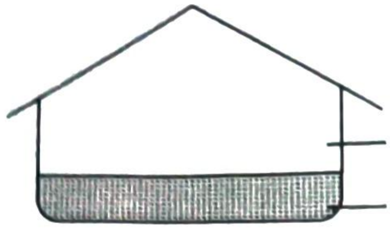
2. Which among the following is responsible for liquification of evaporated water?
(a) *D* (b) *A* (c) *X* (d) *B*

3. *Y* is recharged through the process *X*. *X* is
(a) condensation
(b) evaporation
(c) infiltration
(d) precipitation.

4. Sally used a sealed glass container containing 300 ml of water as a model of the water cycle. She left the model in a garden on a cool night. When she looked at it the next morning, Sally found water droplets on the underside of the cover.



What do the water droplets found o underside of the cover represent i given water cycle?
(a) C (b) *Y* (c) *D* (d) B

5. Which of the following represents the lo water?
(a) *A* (b) *Y* (c) *D* (d) B

24/8/2026

Measurement of Time and Motion

CHAPTER 03

Learning Outcomes...

- Time
- Simple Pendulum
- Motion
- Distance and Speed
- Distance – Time graph
- Velocity – Time graph

Introduction

= could be questions = imp points

Time is the ongoing sequence of events in the past, present, and future that allows us to measure duration and the interval between occurrences. It is a concept that has been essential to human understanding since the beginning of ancient civilization. It is a measure of the duration of events and is commonly defined by the natural phenomena that happens over and over again in a regular fashion. Therefore, day and night were probably the first natural phenomenon to which humans were exposed in a cyclical manner. Thus, day and night were first used to define time. The time interval between the first noon and the second noon was taken for granted as a day. Then humans wanted to know the exact moment of the day, so they invented various timekeeping tools, or clocks. However, these methods were not highly accurate, and it wasn't until the development of electronic oscillators and atomic clocks in the 20th century that timekeeping became much more precise.

Measurement of Time

Time is a phenomenon by which human beings observe and record changes in the environment and the universe.

For example : The time between one sunrise and the next is called a day, a month is measured from one new moon to the next, and a year is fixed as the time taken by the earth to complete one revolution around the sun.

Often we need to measure intervals of time which are much shorter than a day. Clocks or watches are the most common time measuring devices.

One of the earliest means of measurement of time was with the help of movement and change in the position of the sun during the day, and the moon during the night. The position of stars also helped in estimating time during night. Jantar Mantar located in Central Delhi is an advanced form of a Sundial.

Sundial

The device used to measure time is known as a clock. The older clocks like the sand clock, the water clock, were somewhat crude type of clocks.

They only gave an approximation of a duration of time. Sundials were more accurate than the sandclocks and the water clocks. Nowadays, we use a wrist watch or quartz clock. A mechanical

wrist watch is provided with a balance wheel which oscillates like a pendulum. In quartz clo[ck]
fast vibrating crystals of quartz oscillate at a very precise rate and hence measure time more accurately. The atomic clock is the most scientifically developed clock is said to be most accurate of all. The atomic clock is maintained at National Physical Laboratory (NPL) in Delhi, which indicates the standard time. This atomic clock is accurate upto a millionth part of a second.


Water clock


Sand clo[ck]

DO YOU KNOW?

- The study of timekeeping is known as "**horology**", the term clock was used for a **striking clock**, while a clock that did not strike the hours audibly was called a **timepiece**.
- A ghatika is a unit of time measurement in ancient India equal to 24 minutes, where a full day is divided into 60 ghatikas. A ghatika is a unit of time measurement in ancient India equal to 24 minutes, where a full day is divided into 60 ghatikas.

Units of Time

Time refers to the interval between two events taking place.

The SI unit of time is second.

Units of time in decreasing order:

1 millennium	10 centuries
1 century	10 decades
1 decade	10 years
1 year	365 days
1 day	24 hours
1 hour	60 minutes
1 minute	60 seconds

A Simple Pendulum

The famous astronomer Galileo Galilie had performed experiments using a simple pendulum and suggested that it can be used to measure time. He said, if a weight hung from a string is made to swing, it always completes one to-and-fro motion in exactly the same time. In other words, it repeats the oscillation in equal intervals of time.

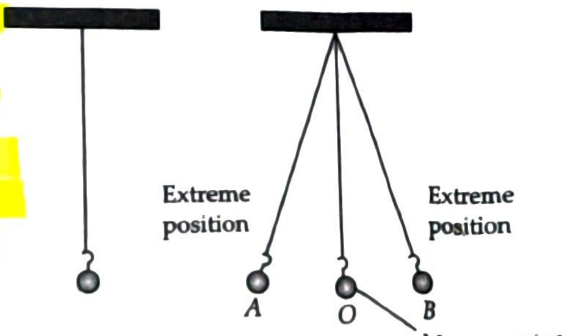


Figure (a) Figure (b)

A simple pendulum consists of a weight (like small metallic ball or a piece of stone) tied to a string. The other end of the string is tied to a rigid support. The metal ball is called the bob.

Figure (a) shows the pendulum at rest in its mean position. When the bob of the pendulum is released after taking it slightly to one side, it begins to move to-and-fro [figure (b)]. The to-and-fro motion of a simple pendulum is an example of periodic and oscillatory motion.

The pendulum is said to have completed one oscillation when its bob is starting from its mean position *O*, moves to *A* to *B* and back to *O*. The pendulum also completes one oscillation when its bob moves from one extreme position *A* to the other extreme position *B* and comes back to *A*. The time taken by the pendulum to complete one oscillation is called its time period. The time period of a simple pendulum depends upon the length of the pendulum and independent of the mass of the bob.

DO YOU KNOW?

The first successful pendulum clock was invented by the Dutch scientist **Christiaan Huygens**, who built it in 1656. While Galileo Galilei had previously discovered the principles of a pendulum's regular swing.

ACTIVITY CORNER

Let us understand the oscillatory motion of a pendulum.

- Make a pendulum by hanging a weight such as a stone from a string, about 1 m in length.

You will also require a stopwatch to measure the time period.

Switch off any fans nearby and let the pendulum come to rest in its mean position. Displace the bob to the right or left by a distance equal to or less than 10% of the length of the pendulum and release it.

Remember that, the time period of a pendulum is constant only for small oscillations.

Start your stopwatch as the bob passes through its mean position.

Allow the bob to swing through 25 complete oscillations and stop the stopwatch at the end of the 25th oscillation.

Divide the time on the stopwatch by 25. This gives the time period of the pendulum.

Increase the length of the string and measure the time period again. Does the time period increase or decrease?

ILLUSTRATIONS

1 Why accurate measurement of time became possible much after accurate measurement of length and mass?

Ans.: Because the measurement of time was done in terms of distance and mass. Mechanical clocks were designed which used to measure time in terms of weight. Accuracy in measuring length and mass enabled us to measure time with more accuracy and precision.

2 What is simple pendulum?

Ans.: A simple device consisting of a small metallic ball and a long thread is called a simple pendulum. The metallic ball is specifically called the bob.

3 What do you mean by time period?

Ans.: Time period is the time taken by an oscillating body to complete one oscillation.

4 Why is time measured, and what is its standard unit?

Ans.: Time measures the duration of events or actions, letting us track when they begin, end, and how long they last. The standard unit of time is seconds, but other units like minutes and hours are commonly used as well, depending on the situation.

Motion

Motion is an important part of our life. We can't perform our day-to-day activities without movement. In order to go to school, office or any other place we need to move either by walking or by car, bus etc. We can't even take food without moving our hands and mouth. Even at the time of sleeping, we breathe air through our lungs and blood flows in arteries and veins. So motion means movement. Motion can also be defined as a continuous change in the position of an object with respect to time. Positions of the object can change in different ways with time so there exist different types of motion and these various types of motion are translatory (linear) motion, vibratory or oscillatory motion, rotatory motion and random motion.

Translatory motion : When an object moves from one point to another along a straight line or a smooth curve, its motion is called translatory. Translatory motion can be of two types- rectilinear and curvilinear, depending on the path followed.

The motion of a body along a straight line with constant or variable speed is called rectilinear motion. For example motion of a car on the straight road.

The motion of a body along a curved path is called curvilinear motion. For example motion of a car in a banked road or in a turn.

Rotatory motion : When an object turns or rotates or spins about an axis, its motion is called rotatory. Motion of earth about its axis and motion of a top about its axis are the examples of rotatory motion.

Oscillatory motion : When a body moves to and fro between two points, repeatedly, its motion is called oscillatory. A child on a swing and the pendulum of a clock are common examples of oscillatory motion. This chapter attempts to build our concept regarding motion and its different aspects.

Slow or Fast Motion

Before understanding slow or fast motion we need to understand the state of rest. An object is said to be at rest if it does not change its position with respect to its surroundings with the passage of time. So, as soon as an object change its state from rest to motion then we can comparatively say that it is either fast or slow motion of that object.

As we know that there are a number of situations in everyday life where we need to understand how slow or fast an object is moving. If you have to travel to a distant city as quickly as possible, you will definitely prefer an aeroplane than a train or a car, for a simple reason that the aeroplane travels faster than a train or a car.

Slow motion

Fast motion

In terms of motion, we can say walking is a slow motion whereas running is a comparatively faster motion.

The distance moved by objects in a given interval of time can therefore help us to decide which object moves faster than the other. We refer to it as speed.

ACTIVITY CORNER

Let's understand slow or fast motion.

- Look at figure (a). It shows the position of some vehicles moving on a road in the same direction at some instant of time.
  Now look at figure (b). It shows the position of the same vehicles after some time.

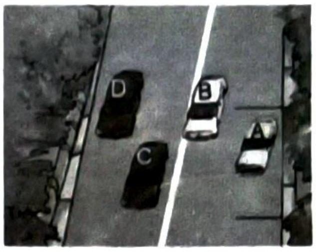


Figure (a)

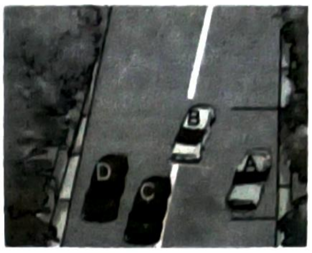


Figure (b)

- From your observation of the two figures, answer the following questions:
- Which vehicle is moving the fastest of all? Which one of them is moving the slowest of all?
- The distance moved by objects in a given interval of time can help us to decide which one is faster or slower.

This activity clearly indicates that to know precisely how fast or slow an object is moving. We need to measure the (i) distance travelled by the object (ii) time taken to travel this distance.

DO YOU KNOW? The sea anemone is considered the slowest animal, moving only a few inches per hour, or about one centimeter per hour. On land, the banana slug is often cited as the slowest, crawling at a pace of about 16.5 cm/min.

Distance

Distance is the measure of the path length between two points, or it is the measure of how far one point is from another.

The SI unit of distance travelled is metre (m) and centimetre (cm) in CGS unit.

Speed

The speed of an object gives us an idea of how slow or fast the object is moving.

Speed is defined as the distance covered by an object in unit time.

The unit for speed is m/s (*i.e.,* metre per second) or km/h (*i.e.,* kilometre per hour).

The formula for calculating the speed of an object is

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$\Rightarrow \frac{1\,\text{km}}{\text{h}} = \frac{1000\ \text{m}}{3600\ \text{s}} = \frac{5}{18}\,\text{m s}^{-1} = \frac{250}{9}\ \text{cm s}^{-1}$$

Learn by heart

DO YOU KNOW? The peregrine falcon is considered to be the fastest bird on the earth. It can achieve a speed of about 390 km/h. The cheetah is the fastest animal with a top speed of about 120 km/h.

Average Speed

The average speed of a body is the total distance travelled by the body divided by the total time taken to cover this distance. While travelling in a car, you may have noticed that it is very difficult to keep the speed of the car uniform because at many places brakes are to be applied to slow down or to stop the car due to various traffic conditions.

$$\text{Average speed} = \frac{\text{Total distance travelled}}{\text{Total time taken to travel that distance}}$$

For example, for a car which travels a distance of 120 km in 3 hours, the average speed is 120/3 = 40 km/h. This does not mean that the car is moving at this speed all the time.

DO YOU KNOW?

- To change km/h into m/s always multiply the given value with 5/18.
  For example, 36 km/h = 36 × 5/18 = 10 m/s
- To change m/s into km/h always multiply the given value with 18/5.
  For example, 20 m/s = 20 × 18/5 = 72 km/h

Measuring Speed

When two bodies move through the same distance, the faster one will take less time.

How do we calculate the speed of a moving body? If a moving body travels a distance d metre in a time of t second, its speed v can be measured as $v = d/t$. If the distance travelled is measured in metre and time in second, then the speed is measured in metre per second (m/s).

Rockets, launching satellites into earth's orbit, often attain speeds up to 8 km/s. On the other hand, a tortoise can move only with a speed of about 8 cm/s. Can you calculate how fast is the rocket compared with the tortoise?

Once you know the speed of an object, you can find the distance moved by it in a given time. All you have to do is to multiply the speed by time. Thus,

Distance covered = Speed × Time

You can also find the time taken by an object to cover a distance while moving with a given speed.

Time taken = Distance/Speed

You might have seen a meter fitted on top of a scooter or a motorcycle. Similarly, meters can be seen on the dashboards of cars, buses and other vehicles. When you observe the dashboard of a car, one of the meters has km/h written at one corner is called a speedometer. It records the speed directly in km/h. There is also another meter that measures the distance moved by the vehicles. This meter is known as an odometer.

DO YOU KNOW? The fastest train in the world is currently the Shanghai Maglev, with an operational speed of up to 460 km/h.

Uniform and Non-uniform Motion

An object moving along a straight line path is said to have uniform motion, if its speed remains constant (*i.e.*, object travels equal distances in equal intervals of time).

The motion of a car moving on a straight road at a constant speed is an example of uniform motion. In uniform motion the average speed is the same as the actual speed.

An object moving along a straight line path is said to have non-uniform motion, if its speed keeps on changing (*i.e.*, object travels unequal distances in equal intervals of time).

The motion of a train starting from a railway station is an example of non-uniform motion.

ILLUSTRATIONS

5 What does the path of an object look like when it is in uniform motion?

Ans.: In uniform motion, the path of an object can be a straight line, curved line, zig-zag line, or even a circle. It can have any shape. This is because in uniform motion, speed is constant. The direction of motion may change.

6 A rubber ball dropped from a certain height bounces to certain height. This height keeps decreasing in subsequent bounces. What type of motion does the ball exhibit?

Ans.: The ball exhibits non-uniform motion.

Displacement

- It is the shortest distance travelled between initial and final positions.
- The displacement of a moving object can be negative, positive or zero but the distance travelled by it will always be a positive value.
- The SI unit of both distance and displacement is metre (m).
- If a body moves between two points along different paths, its displacement remains same for all the paths, though the distance covered by it may be different for the different paths.
  - When an object travel along a path ABC shown in figure (a).

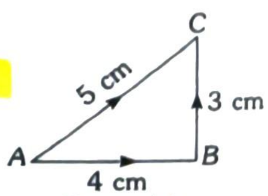


Figure (a)

Distance $= AB + BC$

$= 4 + 3 = 7$ m

COMPETITION WINDOW

Displacement

$= AC = \sqrt{AB^2 + BC^2}$

$= \sqrt{4^2 + 3^2} = 5$ m

- When a body moves from point A to B along a semicircular path shown in figure (b).

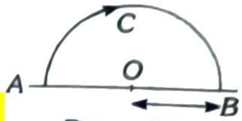
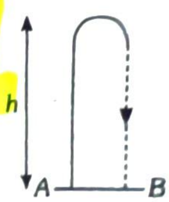


Figure (b)

Distance $ACB = \pi r$

Displacement $AOB = 2r$

- If a body starts moving from a point and returns to the same level, its vertical displacement is zero.



Distance $= h + h = 2h$

Displacement $= 0$.

- Since, displacement is measured from the initial position to the final position, it has a direction. Therefore, it is a quantity that has both magnitude and direction.

Velocity

- When we talk about motion of a body, we should not only know about its speed but also the direction in which the body is moving.
- Velocity is the rate at which an object changes its position. It can be referred as speed in a particular direction. It is defined as the distance travelled by a body in unit time in a given direction.
- Velocity can also be defined as displacement per unit time.

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}}$$

- The SI unit of velocity is same as that of the speed *i.e.*, m s^{-1}.
- Speed is the distance travelled by an obj in unit time and velocity is the distan travelled by an object in unit time in a giv direction. Speed of a moving body can nev be zero but velocity of a moving object w be zero, if it returns to its original positio *i.e.*, when its displacement is zero.

Acceleration

- In non-uniform motion, the speed and th velocity varies with time. The rate at whic velocity varies is called acceleration. Thus, acceleration is the change in veloci per unit time.
- If the velocity of an object varies from a initial value u to the final value v in time then acceleration is given by $a = \frac{v-u}{t}$.
- The S.I. unit of acceleration is m s^{-} or (m/s^2).
- The non-uniform motion is also callec accelerated motion.

ILLUSTRATIONS

7 An express train leaves New Delhi at 6:00 pm and reaches Lucknow at midnight. Its speed is 80 km/h. What is the distance between Delhi and Lucknow?

Ans.: Speed = 80 km/h, time taken = 6 h.
In one hour it travels 80 km. Thus in six hours it travels 80 × 6 = 480 km. Thus, the distance between Delhi and Lucknow is 480 km.

8 Arrange the following speeds in an increasing order:
(a) A bicycle moving with a speed of 18 km/h,
(b) A fast runner running with a speed of 7 m/s,
(c) A car moving with a speed of 2000 m/min.

Ans.: Speeds of different moving bodies can be compared only when all these are expressed in the same unit. Let us express all the speeds in the SI unit, *i.e.*, in m s^{-1}.
(a) Speed of bicycle

$$= \frac{18\text{ km}}{1\text{h}} = \frac{18\times1000\text{m}}{1\times60\times60\text{ s}} = \frac{18000\text{ m}}{3600\text{ s}} = 5\text{ m s}^{-1}$$

(b) Speed of a runner = 7 m s^{-1}
(c) Speed of car

$$= \frac{2000\text{m}}{1\text{min}} = \frac{2000\text{ m}}{1\times60\text{ s}} = 33.3\text{ m s}^{-1}$$

The speeds in the increasing order are 5 m s^{-1}, 7 m s^{-1} and 33.3 m s^{-1}.
This means, the speeds follow the order: Bicycle, Fast runner, Car *i.e.*, 18 km/h, 7 m/s, 2000 m/min (Increasing order)

9 A man riding a scooter travels a distance of 60 metres in 20 seconds. What is the speed of the scooter?

Ans.: $\text{Speed} = \frac{\text{Distance travelled}}{\text{Time taken}}$

Here, distance travelled = 60 m

Time taken = 20 s ; $\text{Speed} = \frac{60\text{ m}}{20\text{ s}} = 3\text{ m s}^{-1}$

Thus, the speed of scooter is 3 metres per second.

Graphs of Motion

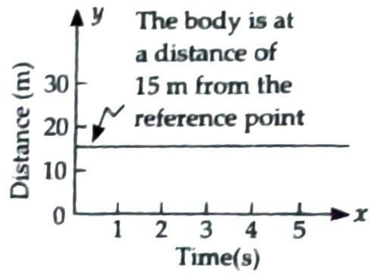
Distance-Time Graph

A moving body changes its position continuously with time. The simplest way to describe the motion of a moving body is to draw its distance-time graph.

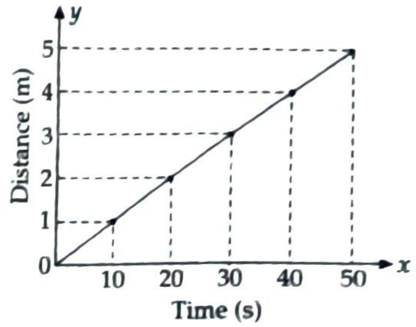
In distance-time graph, the slope is equal to speed of the moving body. Thus, speed of a moving body can be obtained from its distance-time graph.

The distance is always taken along the y-axis and the time is taken along the x-axis in distance-time graph. The distance-time graphs of a body under the following three conditions are described here:

Distance-time graph for a body at rest : When a body does not change its position relative to a fixed reference point, it is said to be in a state of rest. Let us consider a body at a distance of 15 metres from a reference point, if this body is at rest, then it will remain at the same distance at all the times. Therefore, the distance-time graph of a body at rest is a straight line parallel to the time axis (x-axis).

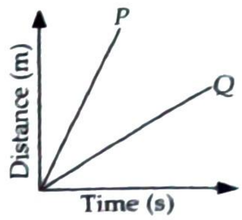


Distance-time graph for a body moving with a uniform speed : When a body covers equal distances in equal intervals of time, it is said to have uniform speed. The distance-time graph of a body which covers a distance of 1 metre in every 10 seconds is shown in figure.



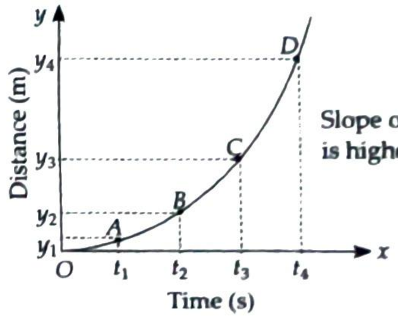
All the points in this graph fall on a straight line, which makes an angle with the x-axis. Thus, the distance-time graph of a body moving with a uniform speed is a straight line making an angle with the x-axis. In other words, the distance travelled by a body moving with uniform speed is directly proportional to time.

A distance-time graph can represent the motion of more than one object. For example, if we consider the uniform motion of two cars P and Q, where P is moving faster than Q, then their motion can be represented by the same distance time graph.



As the speed is equal to the slope of the graph, graph of car P will have a higher slope.

Distance-time graph for a body moving with a non-uniform speed: A body moving with a non-uniform speed covers unequal distances in equal intervals of time. Therefore, the distance-time graph of a body moving with a non-uniform speed is a curve.



Other type of graphs

A bar graph is a diagram which shows information as bars (their rectangles) of different height.

A pie chart is a kind of graph or diagram which shows the percentage composition of 'something' in the form of slices of a circle (the whole circle representing 100 percent).

Displacement-Time Graph

Refer the following data and its graph shown in figure (a)

Displacement	0	2	4	6	4	2	0	–2	–4
Time	0	1	2	3	4	5	6	7	8

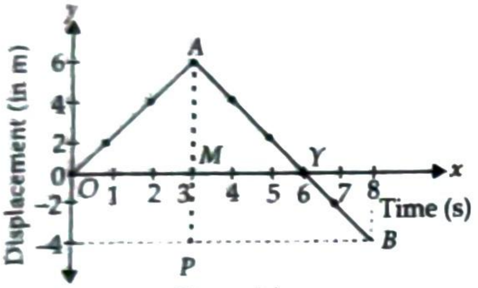


Figure (a)

- The slope of displacement-time graph gives the velocity of the body.

  Slope of $OA = \frac{AM}{OM} = \frac{6\text{ m}}{3\text{ s}} = 2\text{ m s}^{-1}$

  Slope of $AB = \frac{AP}{BP} = -\frac{10\text{ m}}{5\text{s}} = -2\text{ m s}^{-1}$

- The slope of AB is negative because displacement decreases with time. Therefore, velocity during $AB = -2\text{ m s}^{-1}$.
- You will observe three points of difference between displacement-time graph and distance-time graph. All the three points are related to line AB.

(i) Distance can never decrease with time whereas displacement may decrease with time as in AB. So, distance-time graph line can never move downwards whereas displacement-time line can.

(ii) Distance can never be negative, therefore in distance-time graph, you will never find a line below x-axis (when time is taken along x-axis).

But displacement can be negative and therefore, you may find a line below x-axis.

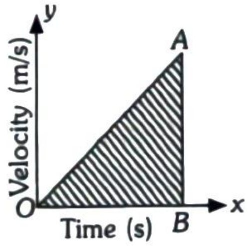
COMPETITION WINDOW

(iii) The slope of a distance-time graph can never be in negative. This is because distance always increases with time, whereas slope of displacement-time graph may be negative.

Velocity-Time Graph

The variation of velocity with time for an object moving in a straight line can be represented by velocity-time graph.

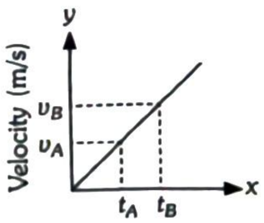
- In this graph, time is taken along x-axis and velocity is taken along y-axis.
- The slope of the velocity-time graph gives the acceleration of the object and the area under the velocity-time curve gives the displacement of the object.



Displacement = Area under the curve OA
= Area of ΔOAB

- Velocity-time graph for a body moving with constant acceleration is,

$$\text{acceleration} = \frac{v_B - v_A}{t_B - t_A}$$



Simple Pendulum

- Time period of a pendulum, $T = 2\pi\sqrt{\frac{L}{g}}$

  Where, L is the effective length of the pendulum and g is the acceleration due to gravity.

ILLUSTRATIONS

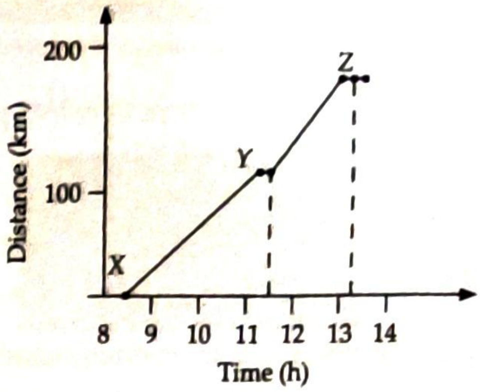
10 The arrival and departure times of a train at three stations, X, Y and Z and the distance of stations Y and Z from station X are given below:

Station	Distance from X(km)	Arrival time(h)	Departure time(h)
X	0	8:00	8:15
Y	120	11:15	11:30
Z	180	13:00	13:15

(a) Plot the distance-time graph for the train, assuming that the motion between the two stations is uniform.

(b) Calculate the uniform speed between the stations X and Y, Y and Z.

Ans.: (a) The distance-time graph for the train is, shown in figure.



(b) Uniform speed between X and Y

$$= \frac{(120-0)\text{ km}}{(11:15-8:15)\text{h}} = \frac{120\text{ km}}{3\text{ h}} = 40\text{ km/h}$$

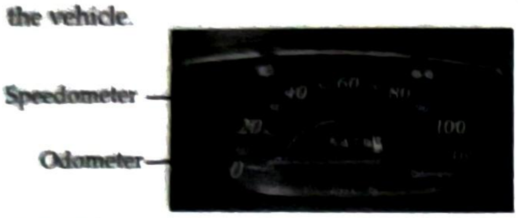
Uniform speed between Y and Z

$$= \frac{(180-120)\text{ km}}{(13:00-11:30)\text{h}} = \frac{60\text{ km}}{1.5\text{ h}} = 40\text{ km/h}$$

CONCEPT MAP

Measuring Speed and Distance

- **Speedometer** : Measures and display the speed of the vehicle in km/h.
- **Odometer** : Measures the distance travelled by the vehicle.



Speed and Velocity

- The speed of an object gives us an idea of how sl or fast the object is moving.
  - Speed is defined as the distance covered by an object in unit time.
  - $\text{Speed} = \dfrac{\text{Distance}}{\text{Time}}$
  - $\text{Average speed} = \dfrac{\text{Total distance travelled}}{\text{Total time taken}}$
- Velocity is the rate at which an object changes i position. It can be referred as speed in a particul direction.
  - Velocity can also be defined as displacement pe unit time.
  - $\text{Velocity} = \dfrac{\text{Displacement}}{\text{Time}}$

Distance and Displacement

- Distance is the measure of the path length between two points, or it is the measure of how far one point is from another.
- Displacement is the shortest distance travelled between initial and final positions.
- The displacement of a moving object can be negative, positive or zero but the distance travelled by it will always be a positive value.

Measurement of Time and Motion

Measurement of Time

Old devices that were used for measuring time are candle clock, sand clock, sundial and water clock.


Sundial


Sand clock


Candle clock

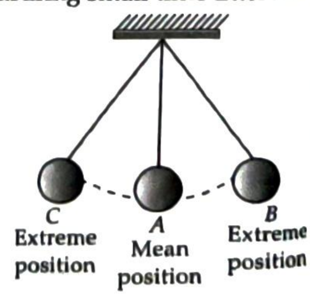

Water clock

Uniform and Non-Uniform Motion

- **Uniform Motion:** A body is said to be in uniform motion if it covers equal distances in equal intervals of time.
- **Non-Uniform Motion:** A body is said to be in non-uniform motion if it covers unequal distances in equal intervals of time

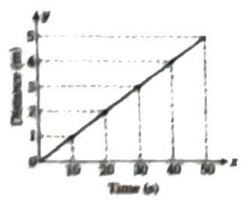
Simple Pendulum

A simple pendulum is used for measuring small time intervals.



Graphical representation of Motion

- Motion of objects can be presented in pictorial form by their distance - time graphs.
- The slope of a distance-time graph indicates the speed of the objects.



Types of Motion

- Translatory
- Rotatory Motion
- Oscillatory Motion

Solved Examples

1. **What is the difference between translatory and rotatory motion? Give their examples.**

Ans.: The motion of an object is said to be translatory, if the position of the object is changing with respect to a fixed point or object with time. For *e.g.*, a car is moving along a road. The motion of an object is said to be rotatory if the motion of all the particles of the body is circular (*i.e.*, along a circular path) with respect to an imaginary line is called the axis of rotation. For *e.g.*, a spinning top, hands of a clock, etc.

2. **Which type of motion is periodic as well as oscillatory?**

Ans.: In case of oscillatory motion, a body as a whole moves back and forth. The motion of a pendulum is an oscillatory motion. Any motion that repeats itself at regular interval of time is called periodic motion. Clock pendulum repeats its motion after a fixed interval of time. So, motion of pendulum is both periodic as well as oscillatory.

3. **How much distance is travelled by an object if it travels at a constant speed of 60 km/h for 2 hours?**

Ans.: Time = 2 hours

Speed = 60 km/h

Distance = ?

$$\text{Speed} = \frac{\text{Distance travelled}}{\text{Time taken}}$$

$$\Rightarrow \quad 60 = \frac{\text{Distance travelled}}{2}$$

Distance travelled = 60 × 2 = 120 km

4. **Find the speed of an object which has travelled 3600 m in 1 minute.**

Ans.: Distance = 3600 m

Time = 1 min = 60 s

Speed = ?

$$\text{Speed} = \frac{\text{Distance travelled}}{\text{Time taken}} = \frac{3600}{60} = 60 \text{ m s}^{-1}$$

5. **The distance between two stations is 360 km. A train takes 8 hours to cover this distance. Calculate the speed of the train.**

Ans.: Distance = 360 km

Time taken = 8 hours

$$\text{Speed} = \frac{\text{Distance covered}}{\text{Time taken}} = \frac{360 \text{ km}}{8 \text{ h}} = 45 \text{ km/h}$$

6. **Can the displacement be greater than the distance travelled by an object? Give reason.**

Ans.: No, the displacement of an object can be either equal to or less than the distance travelled by the object, because the displacement is the shortest distance between initial and final positions of the object while distance travelled is the length of the actual path traversed by the object.

7. **What was a year as per our ancestors?**

Ans.: A year was fixed as the time taken by earth to complete one revolution of the sun.

8. **The odometer of a car reads 1800 km at the start of a trip and 2400 km at the end of the trip. If the trip took 10 h. Calculate the average speed of the car in km/h and m s^{-1}.**

Ans.: Distance covered by the car

s = 2400 – 1800 = 600 km

Trip time = 10 h

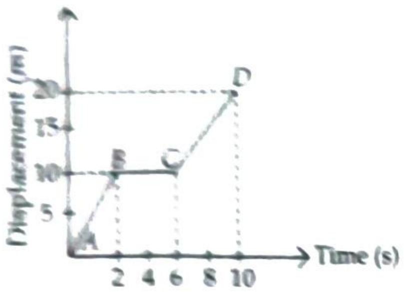
Average speed = ?

(i) $v_{av} = \frac{s}{t} = \frac{600 \text{ km}}{10} = 60 \text{ km/h}$

(ii) In m/s (60 km/h) $= \frac{60 \times 1000}{60 \times 60} = 16.7 \text{ m s}^{-1}$

The average speed of the car in km/h = 60 km/h and in m/s is 16.7 m/s.

9. **This figure represents a displacement-time graph of an object moving along a straight line.**



(a) Describe the motion of the body during the 10 s.

(b) How far did the object travel in the first 2 s?

(c) What was the velocity of the object during the first 2 s?

(d) What was the velocity during the next 4 s?

(e) What was velocity the during the last 4 s ?

Ans.:(a) For part *AB*, the object moved with a uniform velocity, for part *BC*, it was at rest and for part *CD*, it again moved with some other uniform velocity.

(b) In the first 2 s, it has covered 10 m.

(c) Since velocity is given by the slope of the displacement-time graph, *i.e.*,

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}}$$

Velocity during the first 2 s (along *AB*)

$$= \frac{(10-0)}{(2-0)} = 5 \text{ m/s}$$

(d) Velocity during the next 4 s (along *BC*)

$$= \frac{(10-10)}{(6-2)} = 0 \text{ m/s}$$

(e) Velocity during the last 4 s (along *CD*)

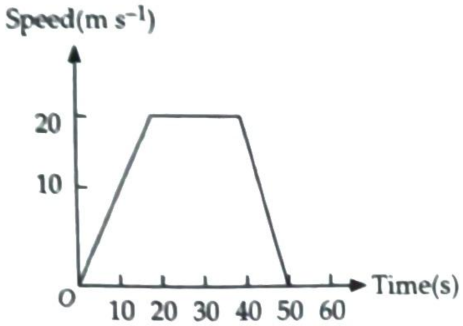
$$= \frac{(20-10)}{(10-6)} = 2.5 \text{ m/s}$$

10. A horse runs a distance of 200 m in 10 s. What is the speed of the horse in (i) m/s (ii) km/h.

Ans.: Speed $= \dfrac{\text{distance}}{\text{time}} = \dfrac{200}{10} = 20 \text{ m/s}$

$$= \frac{200/1000}{10/3600} = \frac{2}{10} \times \frac{3600}{10} = 72 \text{ km/h}$$

11. The graph in figure shows the movement of a car over a period of 50 s. What distance was travelled by the car while its speed was decreasing?

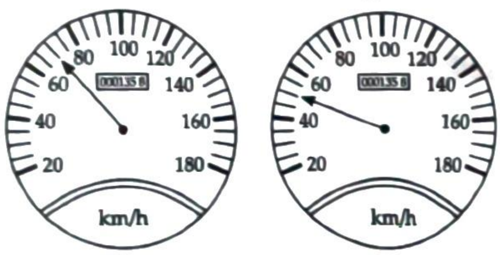


Ans.: The speed was decreasing in the time interval between 40 s and 50 s.

Distance travelled = Area under the graph between 40 s and 50 s

$$= \frac{1}{2} \times 20 \times 10 = 100 \text{ m}$$

12. Look at the two car speedometers drawn below. Which one is travelling faster?



Car 1 Car 2

Ans.: The first car is moving faster than second car.

13. An object travels 16 m in 4 s and then another 16 m in 3 s. What is the average speed of the object?

Ans.: Total distance travelled by the object = 16 m + 16 m = 32 m

Total time taken = 4 s + 3 s = 7 s

$$\therefore \quad \text{Average speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}}$$

$$= \frac{32}{7} = 4.57 \text{ m s}^{-1}$$

$\therefore$ The average speed of the object is 4.57 m s^{-1}.

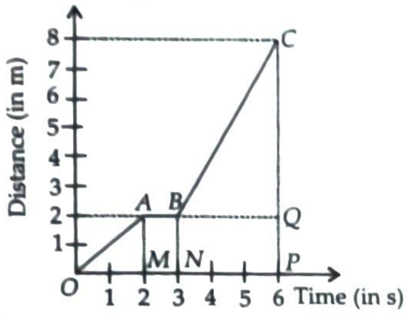
14. The Rajdhani express travels at a speed of 80 km/h. The distance between Delhi and Chennai is 2200 km. If it starts from Delhi at 3.00 p.m., then at what time does it reach Chennai?

Ans.: Speed = 80 km/h; Distance = 2200 km.

It covers 80 km in one hour. Therefore, it covers 2200 km in (2200 ÷ 80) = 27.5 hours.

It reaches Chennai next day at 6.30 p.m.

15. Interpret the given distance-time graph.



Ans.: For O to A, when we get a straight line inclined to time axis in a distance-time graph, it is case of constant speed. The speed is the slope of the line.

$$\text{Speed} = \frac{AM}{OM} = \frac{2\text{ m}}{2\text{ s}} = 1\text{ m s}^{-1}$$

For A to B, when we get a straight line parallel to time axis in a distance-time graph, the body is at rest.

For B to C, it is a case of constant speed where

$$\text{Speed} = \frac{CQ}{BQ} = \frac{6\text{ m}}{3\text{ s}} = 2\text{m s}^{-1}$$

16. A spaceship travels 36,000 km in one hour. Express its speed in km/s. *(NCERT Exemplar)*

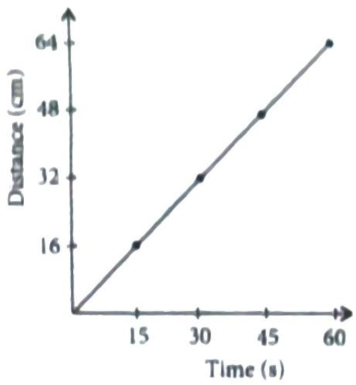
Ans.: Speed of spaceship = 36,000 km/h

$$= \frac{36000\text{ km}}{60 \times 60\text{s}} = 10\text{ km/s}$$

17. Plot a distance-time graph of the tip of the second hand of a clock by selecting 4 points on x-axis and y-axis respectively. The circumference of the circle traced by the second hand is 64 cm. *(NCERT Exemplar)*

Ans.:

Time (s)	x	15	30	45	60
Distance (cm)	y	16	32	48	64



18. If Boojho covers a certain distance in one hour and Paheli covers the same distance in two hours, who travels in a higher speed? *(NCERT Exemplar)*

Ans.: Boojho moves at a higher speed as he covers the same distance in a lesser time than Paheli.

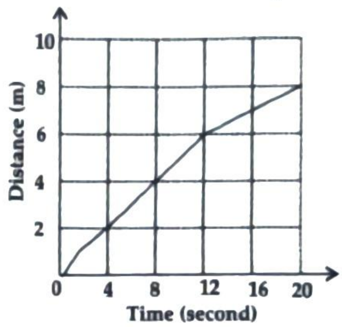
19. The average age of children of Class VII is 12 years and 3 months. Express this age in seconds. *(NCERT Exemplar)*

Ans.: 12 years 3 months

= 12 × 365 + 3 × 30 = 4470 days

= 4470 × 24 × 60 × 60 s = 386208000 s

20. Figure given below is the distance-time graph of the motion of an object.



(i) **What will be the position of the object at 20 s?**

(ii) **What will be the distance travelled by the object in 12 s?**

(iii) **What is the average speed of the object?** *(NCERT Exemplar)*

Ans.: (i) At 20 s, the object will be 8 m away from the starting point.

(ii) In 12 s, distance travelled by the object will be 6 m.

(iii) Average speed of the object

$$= \frac{\text{Total distance}}{\text{Time taken}} = \frac{8\text{ m}}{20\text{ s}} = 0.4\text{ m/s}$$

NCERT Section

1. Calculate the speed of a car that travels 150 metres in 10 seconds. Express your answer in km/h.

2. A runner completes 400 metres in 50 seconds. Another runner completes the same distance in 45 seconds. Who has a greater speed and by how much?

3. A train travels at a speed of 25 m/s and covers a distance of 360 km. How much time does it take?

4. A train travels 180 km in 3 h. Find its speed in:
   (i) km/h
   (ii) m/s
   (iii) What distance will it travel in 4h if it maintains the same speed throughout the journey?

5. The fastest galloping horse can reach the speed of approximately 18 m/s. How does this compare to the speed of a train moving at 72 km/h?

6. Distinguish between uniform and non-uniform motion using the example of a car moving on a straight highway with no traffic and a car moving in city traffic.

7. Data for an object covering distances in different intervals of time are given in the following table. If the object is in uniform motion, fill in the gaps in the table.

Time (s)	Distance (m)
0	0
10	8
20	
30	24
	32
50	40
70	56

8. Car covers 60 km in the first hour, 70 km in the second hour, and 50 km in the third hour. Is the motion uniform? Justify your answer. Find the average speed of car.

9. Which type of motion is more common in daily life—uniform or non-uniform? Provide three examples from your experience to support your answer.

10. Data for the motion of an object are given in the following table. State whether the speed of the object is uniform or non-uniform. Find the average speed.

Time (s)	0	10	20	30	40	50	60	70	80	90	100
Distance (m)	0	6	10	16	21	29	35	42	45	55	60

11. A vehicle moves along a straight line and covers a distance of 2 km. In the first 500 m, it moves with a speed of 10 m/s and in the next 500 m, it moves with a speed of 5 m/s. With what speed should it move the remaining distance so that the journey is complete in 200 s? What is the average speed of the vehicle for the entire journey?

Exercise

Multiple Choice Questions

LEVEL - 1

Measurement of Time and Simple Pendulum

1. Which of the following device was not used by our ancestors to measure time?
   (a) Sand clock (b) Quartz clock
   (c) Sundial (d) Water clock
   (NCERT Page 106) R

2. Which of the following cannot be used for measurement of time?
   (a) A leaking tap.
   (b) Simple pendulum.
   (c) Shadow of an object during the day.
   (d) Blinking of eyes. *(NCERT Exemplar)*

3. Which of the following does not show periodic motion?
   (a) Pendulum
   (b) Hands of a clock
   (c) Revolution of Earth around the Sun
   (d) A car moving at a constant speed
   (Page 109) R

4. How many minutes are there in 6 hours?
   (a) 120 (b) 360 (c) 180 (d) 3600
   (Page 111) Ap

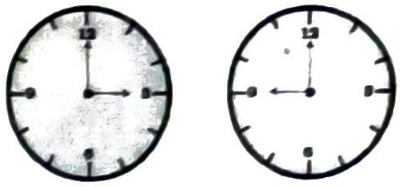
5. On Saturday night, Prachi spent 18 minutes on social science homework, 35 minutes on mathematics homework and 22 minutes on english homework. How much time did she spend on her homework?
   (a) 1 h 75 min (b) 2 h 15 min
   (c) 2 h 75 min (d) 1 h 15 min
   (Page 111) Ap

6. How many weeks are there in 56 days?
   (a) 8 (b) 7
   (c) 6 (d) 9 *(Page 111)* Ap

7. What part of second is called nanosecond?
   (a) One hundredth
   (b) One thousandth
   (c) One millionth
   (d) One billionth *(Page 111)* R

8. How many minutes are there in 4.5 hours?
   (a) 290 (b) 240 (c) 270 (d) 450
   (Page 111) Ap

9. Two clocks *A* and *B* are shown in figure. Clock *A* has an hour and a minute hand, whereas clock *B* has an hour hand, minute hand as well as a second hand. Which of the following statement is correct for these clocks?



(A) (B)

   (a) A time interval of 30 seconds can be measured by clock *A*.
   (b) A time interval of 30 seconds cannot be measured by clock *B*.
   (c) Time interval of 5 minutes can be measured by both *A* and *B*.
   (d) Time interval of 4 minutes 10 seconds can be measured by clock *A*.
   (NCERT Exemplar)

Motion

10. Change in position of an object with respect to time is known as
    (a) speed (b) distance
    (c) motion (d) time.
    (Page 113) R

11. Train *A* travels with a speed of 80 km/h from Delhi to Chennai. Train *B* travels from Delhi to Chennai with a speed of 75 km/h.

Which of the following statements is true?

(a) Trains A and B have same velocity as they are travelling in the same direction.

(b) Trains A and B have same velocity but different speed.

(c) Trains A and B have different speeds and different velocities.

(d) Can't say, the given information is not enough. *(Page 113)* U

12. If we denote speed by S, distance by D and time by T, the relationship between these quantities is

(a) $S = D \times T$ (b) $T = \frac{S}{D}$

(c) $S = \frac{1}{T} \times D$ (d) $S = \frac{T}{D}$

(NCERT Exemplar)

13. A car travels with speed of 50 km/h in the 1st hour and 100 km/h in the 2nd hour. What is the total distance covered by the car in the two hours?

(a) 50 km (b) 100 km

(c) 150 km (d) 200 km

(Page 115) Ap

14. Riaz takes 20 min to travel to his school with a speed of 3 m/s. How far is the school?

(a) 2 km (b) 3.2 km

(c) 3.6 km (d) 4.1 km

(Page 115) Ap

15. With what speed should a car travel so that it can cover a distance of 5 km in 5 min?

(a) 1 km/h (b) 5 km/h

(c) 12 km/h (d) 60 km/h

(Page 115) Ap

16. A jet is moving with a speed of 180 km/h. What is its speed in m/s?

(a) 10 m/s (b) 50 m/s

(c) 200 m/s (d) 500 m/s

(Page 115) Ap

17. Which of the following is used to measure the distance travelled by a moving vehicle?

(a) Speedometer (b) Kilometer

(c) Meter (d) Odometer

(Page 116) R

18. Motion of a car on a straight path is an example of

(a) rectilinear motion (b) circular motion

(c) rotatory motion (d) periodic motion

(Page 116) R

19. Which of the following shows a motion in a straight line?

(a) Train moving on a straight bridge

(b) Tree swaying

(c) Child on a merry-go-round

(d) None of these. *(Page 116)* R

20. Which of the following is not in uniform motion?

(a) A bullet travelling with a constant speed in a straight line.

(b) A swinging pendulum.

(c) The rotation of earth.

(d) Both (b) and (c). *(Page 116)* U

21. Which of the following is not matched correctly?

(a) Speedometer : Speed

(b) Odometer : Odour

(c) Anemometer : Wind speed

(d) Stopwatch : Time *(Page 116)* U

22. When a body covers equal distance in equal interval of time, it is said to be in

(a) rotational motion

(b) uniform motion

(c) non-uniform motion

(d) oscillatory motion *(Page 117)* R

23. Which one of the following is true for the uniform speed?

(a) If speed of the object varies at different interval of time, the object has uniform speed.

(b) If speed of the object is constant at different intervals of time, the object has uniform speed.

(c) If a moving object suddenly comes to rest, the object has uniform speed.

(d) None of these. *(Page 117)* U

24. A cyclist covers 25 km in first hour, 30 km in the second hour and 35 km in the third hour. The cyclist is in

(a) uniform motion
(b) non-uniform motion
(c) can't say if the body is in uniform motion or in non-uniform motion
(d) none of these. *(Page 117)* Ap

25. A body is in uniform motion. On which type of path should it travel so that its motion can be called periodic motion?
(a) Straight path (b) Curved path
(c) Circular path (d) None of these. *(Page 117)* U

26. When a body spins, the body is in
(a) rectilinear motion (b) circular motion
(c) rotatory motion (d) periodic motion R

27. Motion of Mars around the Sun is an example of
(a) rectilinear motion (b) oscillatory motion
(c) periodic motion (d) rotatory motion U

28. Which of the following does not show oscillatory motion?
(a) Swing (b) Fan
(c) See-saw (d) Pendulum R

Graphs of Motion

29. The speed of an object can be determined from a graph is called
(a) work-energy graph
(b) force-distance graph
(c) distance-time graph
(d) none of these. R

30. Look at the following graph.

The above graph shows that the object is
(a) at rest
(b) in oscillatory motion
(c) in uniform motion
(d) in uniform acceleration Ap

LEVEL - 2

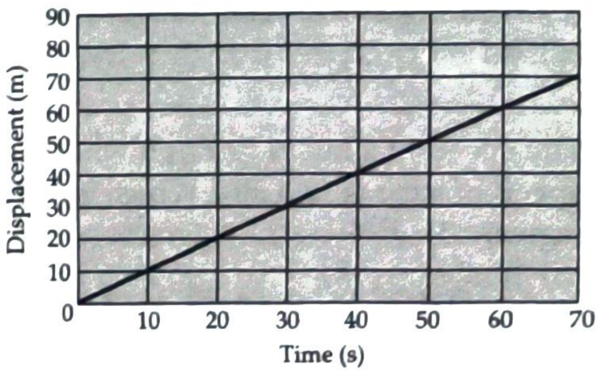
31. How many years are there in a millennium?
(a) 10 (b) 100
(c) 1000 (d) 10000 Ap

32. In a long distance race, the athletes were expected to take four rounds of the track such that the line of finish was same as the line of start. Suppose the length of the track was 200 m. Then what is the displacement of the athletes when they touch the finish line?
(a) zero (b) 3 m
(c) 5 m (d) 7 m Ap

33. If a car covers 2/5th of the total distance with v_1 speed and 3/5th distance with v_2, then average speed is
(a) $\frac{1}{2}\sqrt{v_1 v_2}$ (b) $\frac{v_1+v_2}{2}$
(c) $\frac{2v_1v_2}{v_1+v_2}$ (d) $\frac{5v_1v_2}{3v_1+2v_2}$ Ap

34. A body goes 30 km east then 20 km north then 80 km west and then 50 km east. Find the displacement of the body from the initial position.
(a) 0 km (b) 20 km
(c) 50 km (d) 180 km Ap

35. Look at the following graph carefully and find the velocity of the moving body.



(a) 0 m/s (b) 1 m/s
(c) 2 m/s (d) 3 m/s

36. Boojho walks to his school which is at a distance of 3 km from his home in 30 minutes. On reaching he finds that the school is closed and comes back by a

bicycle with his friend and reaches home in 20 minutes. His average speed in km/h is

(a) 8.3 (b) 7.2 (c) 5 (d) 3.6

(NCERT Exemplar)

37. When a body is moving with a constant velocity, the displacement time graph will be a

(a) Straight line (b) Curved line
(c) Parabola (d) None of these

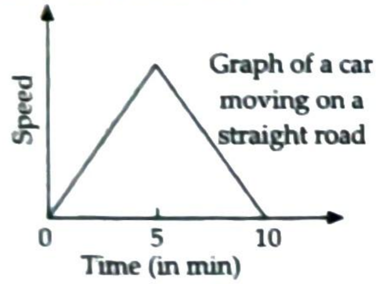
U

38. "When a body is moving with a constant velocity, the velocity-time graph is a straight line parallel to time axis". The given statement is

(a) true (b) false
(c) partly true (d) none of these

An

39. According to the following graph, what happens to the distance covered by the body from 0-10 minutes?



(a) It goes on increasing.
(b) It goes on decreasing.
(c) It first increases and then decreases.
(d) It first decreases and then increases.

An

40. The speed of a bus is 72 km/h whereas the speed of a car is 12.5 m/s. Which is moving faster?

(a) Bus
(b) Car
(c) Both with same speed
(d) Data insufficient.

An

41. Which one of the following is an example of uniform motion?

(a) A train entering a railway platform.
(b) An aeroplane flying at a speed of 600 km/h.
(c) A bee going from one flower to another collecting nectar.
(d) A boy cycling in a crowded market.

U

42. A train travels at 60 km/h for 0.52 h; at 30 km/h for the next 0.24 h and at 70 km/h for the next 0.71 h. What is the average speed of the train?

(a) 12.5 km/h (b) 10 km/h
(c) 20 km/h (d) 59.9 km/h

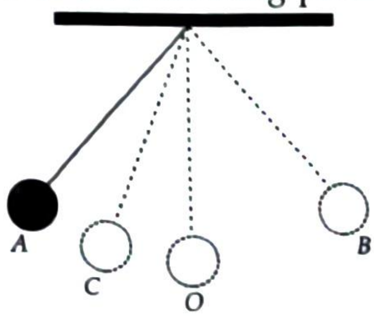
Ap

43. Train *A* travels with a speed of 90 km/h from Hyderabad to Bangalore. Train *B* travels with a speed of 90 km/h from Bangalore to Hyderabad.
Which of the following statements is true?

(a) Both trains are travelling with the same speed and same velocity.
(b) Both trains have the same velocity.
(c) Trains *A* and *B* have the same speed but different velocities.
(d) Trains *A* and *B* have the same velocity but different speeds.

An

44. Figure shows an oscillating pendulum.



Time taken by the bob to move from *A* to *C* is t_1 and from *C* to *O* is t_2. The time period of this simple pendulum is

(a) $(t_1 + t_2)$ (b) $2 (t_1 + t_2)$
(c) $3 (t_1 + t_2)$ (d) $4 (t_1 + t_2)$

(NCERT Exemplar)

45. The bus decreases its speed from 80 km h^{-1} to 60 km h^{-1} in 5 s. The acceleration of the bus is

(a) 0.1 m/s^2 (b) –1.11 m/s^2
(c) –2.1 m/s^2 (d) zero

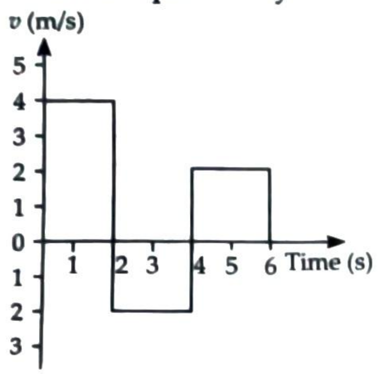
Ap

46. A train is travelling from west to east at an average speed of 120 km/h. How far does this train travel in 6.0 second?

(a) 900 m (b) 500 m
(c) 200 m (d) 20 m Ap

47. A man walks on a straight road from his home to market 2.5 km away with a speed of 5 km/h. Finding the market closed, he instantly turns and walks back home with a speed of 7.5 km/h. The average speed of the man over the interval of time 0 to 40 min is equal to
(a) 9.1 km/h (b) 6.25 km/h
(c) 7.5 km/h (d) 5.62 km/h Ap

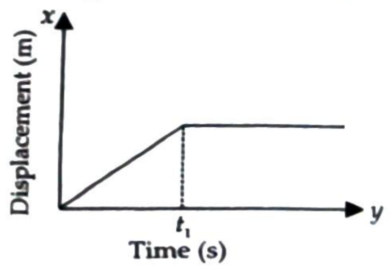
48. The velocity-time graph of a body moving in a straight line is shown in the figure. The displacement and distance travelled by the body in 6 s are respectively



(a) 8 m, 16 m (b) 16 m, 8 m
(c) 16 m, 16 m (d) 8 m, 8 m. An

49. A ship is moving at a speed of 56 km/h. One second later it is moving at 58 km/h. What is its acceleration ?
(a) 7200 km/h^2 (b) 3295 km/h^2
(c) 5490 km/h^2 (d) 8200 km/h^2 Ap

50. The figure shows the displacement-time graph of a particle moving along the x-axis. Which of the following statement will describe the motion of the particle correctly?



(a) The particle is moving continuously.
(b) The particle is at rest.
(c) The velocity of the particle increases upto time t_1 and then becomes constant.
(d) The particle moves with a constant velocity upto time t_1 and then stops.
An

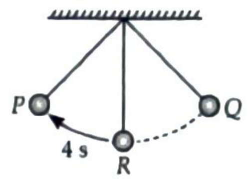
LEVEL - 3 (HOTS)

Competency Focused Questions (CFQs)

51. The second's hand of a watch is 2 cm long. The speed of the tip of second's hand is
(a) 0.21 cm s^{-1} (b) 2.1 cm s^{-1}
(c) 21 cm s^{-1} (d) 0.021 cm s^{-1}

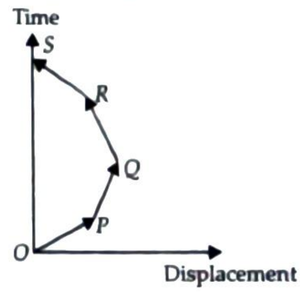
52. A train leaves a station X at 5:00 pm and reaches another station Y at midnight. Its speed is 90 km h^{-1}. What is the distance between X and Y?
(a) 480 km (b) 630 km
(c) 800 km (d) 830 km

53. If the time taken by the bob of a pendulum to move from point R to P is 4 s, then the time taken by it to complete its one oscillation is



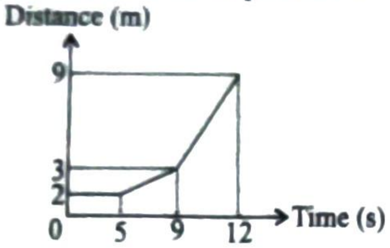
(a) 8 s (b) 12 s
(c) 16 s (d) 4 s

54. Which of the following options is correct for the object having a straight line motion represented by the graph shown in figure?



(a) The object moves with constantly increasing velocity from O to P and then it moves with constant velocity.
(b) Velocity of the object increases uniformly.
(c) Average velocity is zero.
(d) The graph shown is impossible.

55. The distance-time graph of motion of a cycle is shown here. What is the average speed of the cycle over a time period of 12 s ?



(a) 0.48 m s^{-1} (b) 0.58 m s^{-1}
(c) 2.17 m s^{-1} (d) 3.17 m s^{-1}

56. A train moving on linear path travels a distance D at a constant speed of 30 km/h, then it travels the same distance in opposite direction and reaches initial position at a constant speed of 45 km/h. What is the average speed of the train?

(a) 36 km/h (b) 10 km/h
(c) 0 (d) 75 km/h

57. Rohan is going to school with his father by car. He decides to note the readings on the odometer of the car after every five minutes till he reaches the school. The given table shows the odometer readings of his journey.

Time (a.m.)	7:30	7:35	7:40	7:45	7:50
Odometer reading (km)	36593	36596	36599	36602	36605

If he left for school at 7:30 am, then find the average speed of the car.

(a) 5 m/s (b) 6 m/s
(c) 10 m/s (d) 15 m/s

58. Read the given statements and select the correct option.

Statement 1 : Circular motion is the motion along a curve traced out by a moving object such that its distance from a fixed point is constant.

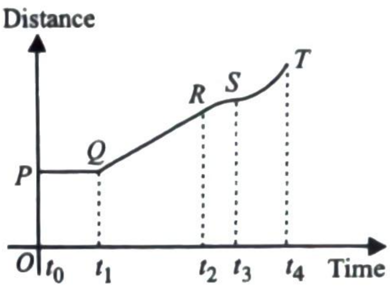
Statement 2 : The motion of a wheel of a moving bicycle is same as the motion of a mark on the blades of a moving electric fan.

(a) Both statements 1 and 2 are true and statement 2 is the correct explanation of statement 1.
(b) Both statements 1 and 2 are true but statement 2 is not the correct explanation of statement 1.
(c) Statement 1 is true but statement 2 is false.
(d) Statement 1 is false but statement 2 is true.

59. A person is running along a circular track of area 625 π m^2 (π = 22/7) with a constant speed. Find the displacement in 15 seconds if he has to complete the race in 30 seconds.

(a) 200 m (b) 100 m
(c) 25 m (d) 50 m

60. Distance-time graph of an object is shown in the figure. Object is at rest in a time interval from



(a) t_3 to t_4 (b) t_2 to t_3
(c) t_1 to t_2 (d) t_0 to t_1

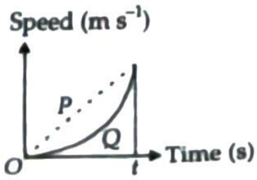
61. A car covers 30 km at a uniform speed of 60 km/h and the next 30 km at a uniform speed of 40 km/h. The total time taken is

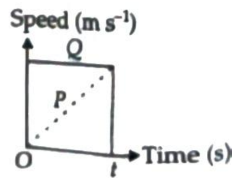
(a) 30 min (b) 45 min
(c) 75 min (d) 120 min

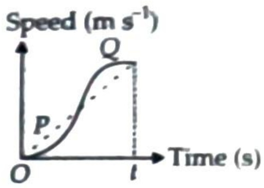
62. An athlete completes one round of a circular track of radius R in 40 seconds. The displacement at the end of 2 minute 20 second will be

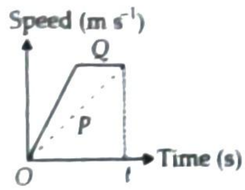
(a) zero (b) $2R$
(c) πR (d) $7\pi R$

63. In a 200 m race, two competitors P and Q reached the finishing line at the same time. Which of the following graphs shows that both the competitors reached the finishing line at time t?

(a)


(b)


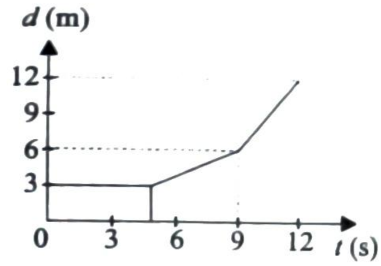
(c) 

(d) 

64. Parsec is the unit of

(a) distance (b) time
(c) velocity (d) angle

65. The distance-time graph of a motorcycle is shown in the figure. When was the motorcycle travelling at a speed of 2 m s^{-1}?

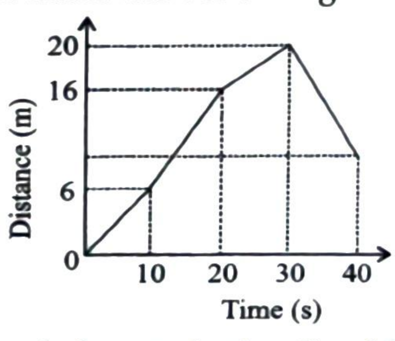


(a) 2.5 s (b) 5.0 s
(c) 8.0 s (d) 10.5 s

66. The wrong unit conversion among the following is

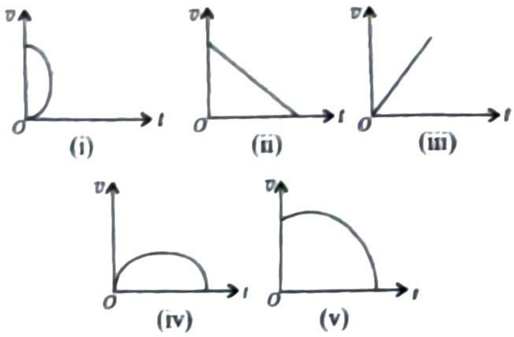
(a) 1 day – 24 hours
(b) 1 year – 12 months
(c) 1 week – 7 days
(d) 1 hour – 360 seconds

67. Study the distance-time graph of a car, given in the figure and choose the correct statement from the following.



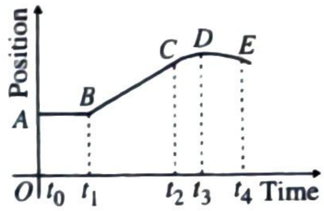
(a) The car is fastest during first 10 seconds.
(b) The car is slowest in between 10 and 20 seconds.
(c) Distance travelled during last 10 seconds is 4 m.
(d) Distance travelled between 10 s to 30 s is 14 m.

68. Which of the following cannot be speed-time ($v - t$) graph(s)?



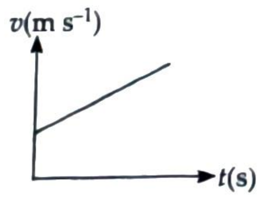
(a) (ii) and (iv) only (b) (iii) and (v) only
(c) (iv) only (d) (i) only

69. Position – time graph of an object is shown in the figure. Object moves with constant speed in time interval from



(a) t_3 to t_4 (b) t_2 to t_3
(c) t_1 to t_2 (d) t_0 to t_1

70. A speed-time (v-t) graph of an object is shown in the figure.



Which of the following best describes the graph?

(a) The object drops upward through vacuum.
(b) The object drops through air.
(c) The object is being thrown downwards through vacuum.
(d) The object is being thrown downwards through air.

Fill in the Blanks

1. ________ is a motion in which a body repeats the motion at equal intervals of time.
2. The time taken by a pendulum to complete one oscillation is called its ________.

3. When the distance travelled by a body at fixed intervals of time is not the same, it is said to be in ________.

4. To ascertain the distance travelled by the vehicle, an ________ is fitted along with the speedometer.

5. The basic unit of time is ________.

6. The distance-time graph for the motion of an object moving with a constant speed is a ________.

7. The distance covered by an object in a unit time is known as the ________ of the object.

8. The ages of stars and planets are often expressed in ________ of years.

9. Distance covered = ________ × time.

10. Motion of objects can be represented in pictorial form by their ________.

True or False

1. A sundial works by casting a shadow in different positions at different times of day.
2. The time required to complete one oscillation depends on the mass of the string of simple pendulum.
3. The direction of motion and the speed of a body in uniform motion do not change.
4. A pendulum clock is much more accurate than a quartz clock.
5. The graph of distance versus time is a straight line for non-uniform motion.
6. The distance between New Delhi to Lucknow is measured in metre.
7. Anything that occurs or appears at regular intervals can help us to measure time.
8. A year was measured as the time taken by the moon to complete one revolution around the sun.
9. Speed of a moving body can never be zero.
10. The speedometer in vehicle gives the speed of the moving vehicle directly in km/h.

Match the Following

In this section, each question has two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (a), (b), (c) and (d) out of which one is correct.

1.

	List-I		List-II
(P)	When a body changes its position with time	(1)	Non-uniform motion
(Q)	Interval between two events	(2)	Motion
(R)	SI unit of speed	(3)	Time
(S)	Unequal distance in equal intervals of time	(4)	Metre per second

(a) (P)-(3), (Q)-(4), (R)-(1), (S)-(2)
(b) (P)-(3), (Q)-(1), (R)-(2), (S)-(4)
(c) (P)-(2), (Q)-(3), (R)-(4), (S)-(1)
(d) (P)-(2), (Q)-(4), (R)-(3), (S)-(1)

2.

	List-I		List-II
(P)	Day	(1)	Second
(Q)	Month	(2)	Metre
(R)	Time	(3)	The time from one sunrise to the next
(S)	Distance	(4)	One new moon to the next

(a) (P)-(3), (Q)-(4), (R)-(1), (S)-(2)
(b) (P)-(4), (Q)-(1), (R)-(2), (S)-(3)
(c) (P)-(1), (Q)-(4), (R)-(3), (S)-(2)
(d) (P)-(1), (Q)-(3), (R)-(2), (S)-(4)

Assertion & Reason Type

Directions : In the following questions, a statement of assertion is followed by a statement of reason. Mark the correct choice as :

(a) If both assertion and reason are true and reason is the correct explanation of assertion
(b) If both assertion and reason are true but reason is not the correct explanation of assertion
(c) If assertion is true but reason is false
(d) If assertion is false but reason is true.

1. **Assertion** : Displacement of a body may be zero when distance travelled by it is not zero.

Reason : The displacement is the longest distance between initial and final position.

2. **Assertion** : Position-time graph of a stationary object is a straight line parallel to time axis.

Reason : For a stationary object, position does not change with time.

3. **Assertion** : The speedometer of a car measures the instantaneous speed of the car.

Reason : Average speed is equal to the total distance covered by an object divided by the total time taken.

4. **Assertion** : The position-time graph of uniform motion of a body can have negative slope.

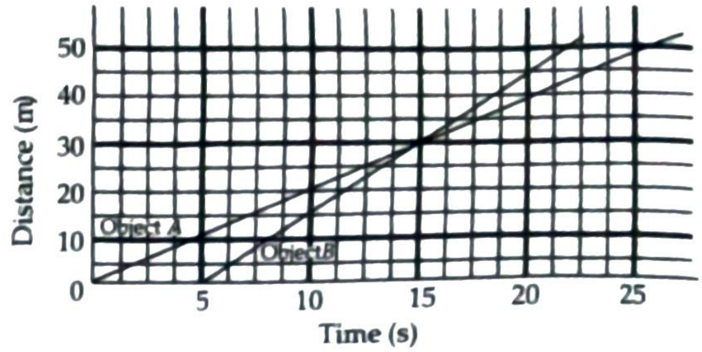
Reason : When the speed of body decreases with time, the position-time graph of the moving body has negative slope.

5. **Assertion** : The average speed of a body may be equal to the arithmetic mean of the speeds during its motion if time intervals are equal.

Reason : Average speed is equal to the total distance travelled divided by the total time taken.

Comprehension Type

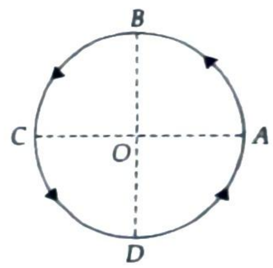
PASSAGE-I : The distance-time graph is a line graph, where the distance travelled by the body is plotted against time. The distance-time graph can represent the motion of more than one object. Graphical representation of motion of two different objects A and B has been shown in the graph. Study the graph and answer the following questions.



1. Both the objects are in
   (a) Uniform motion
   (b) Non-uniform motion
   (c) Periodic motion
   (d) Circular motion

2. Which one of the two objects have greater speed?
   (a) Object A
   (b) Object B
   (c) Both the objects have equal speed
   (d) None of these

PASSAGE-II : Displacement is the shortest distance between two points. It can also be stated as the change in the position of a moving body in a particular direction. Consider a particle moving on a circular path of radius R as shown in the figure. Study the figure and answer the following questions.



1. The distance travelled and displacement of the particle from A to B respectively are
   (a) $\frac{\pi R}{4}, R$ (b) $\frac{\pi R}{2}, 2R$
   (c) $\frac{\pi R}{4}, \sqrt{2}R$ (d) $\frac{\pi R}{2}, \sqrt{2}R$

2. The distance travelled and displacement of the particle from A to C respectively are
   (a) $\frac{\pi R}{2}$, $2R$ (b) πR, $2R$
   (c) $\frac{\pi R}{2}, \frac{\pi R}{2}$ (d) πR, πR

Subjective Problems

Very Short Answer Type

1. Give the name of device which measures time.
2. Give the term used for the time taken by a pendulum to complete one oscillation.
3. What is the shape of distance-time graph for a body moving with uniform motion?
4. Define speed.
5. Name the device which records the speed of a moving vehicle.
6. What type of motion is possessed by a body, when it moves unequal distances in equal intervals of time or vice-versa?
7. Given one example of periodic motion.
8. Convert 54 km/h into m/s.
9. What do you mean by zero speed?
10. Give example of any situation showing translatory motion.
11. Define velocity.
12. What is odometer?
13. With the help of distance-time graph show that the object is stationary.
14. What is rectilinear motion?
15. Give an example of non-uniform motion.
16. A simple pendulum takes 32 s to complete 20 oscillations. What is the time period of the pendulum?
17. The distance between two stations is 240 km. A train takes 4 hours to cover this distance. Calculate the speed of the train.
18. Salma takes 15 minutes from her house to reach her school on a bicycle. If the bicycle has a speed of 2 m/s, calculate the distance between her house and the school.
19. Explain the limitations of a sundial.
20. On what principle are modern watches based?

Short Answer Type

1. Explain the following terms with respect to a pendulum: time period, oscillation and mean position.
2. A car is moving on a hill station and covers the distance as per the data given. Read the following data and construct a distance-time graph on a graph paper.

Time (in minutes)	Distance (in km) from the starting point
10	5
20	10
30	15
40	20
50	25

3. Which type of watches are used nowadays to measure time? Explain one of them.
4. Define the following terms:
   (a) Non-uniform motion
   (b) Average speed.
5. A car covers a distance of 60 km in the first hour, 55 km in the second hour, 70 km in the third hour, 50 km in the fourth hour, 65 km in the fifth hour and 60 km in the sixth hour. Find the average speed of the car and plot a distance-time graph for the same.
6. Explain uniform and non-uniform motion.
7. Explain how Galileo contributed to the development of clocks.
8. A car travels a distance of 270 kilometer in $4\frac{1}{2}$ hours. Find its speed.
9. Draw a distance-time graph for non-uniform motion.
10. The odometer of a car reads 57321.0 km when the clock shows the time 08:30 AM. What is the distance moved by the car, if at 08:50 AM, the odometer reading has changed

to 57336.0 km? Calculate the speed of the car in km/min during this time. Express the speed in km/h also.

Long Answer Type

1. Explain sundial, water clock and sand clock with diagram.
2. What is simple pendulum? Explain its working.
3. Explain with the example how can we choose a suitable scale.
4. There are bodies in motion which execute two or more types of motion at the same time. Give two examples and explain.
5. The following is the distance-time table of a moving car.

Time	Distance
10.05 am	0 km
10.25 am	5 km
10.40 am	12 km
10.50 am	22 km
11.00 am	26 km
11.10 am	28 km
11.25 am	38 km
11.40 am	42 km

(a) Use a graph paper to plot the distance travelled by the car versus the time.

(b) When was the car travelling at the greatest speed?

(c) What is the average speed of the car?

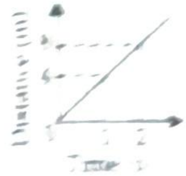
(d) What is the speed between 11.25 am and 11.40 am?

(e) During a part of the journey, the car was forced to slow down to 12 km/h. At what distance did this happen?

Integer/Numerical Value Type

1. An electric train is moving with a velocity of 120 km/h. How much distance (in km) will it move in 30 s?

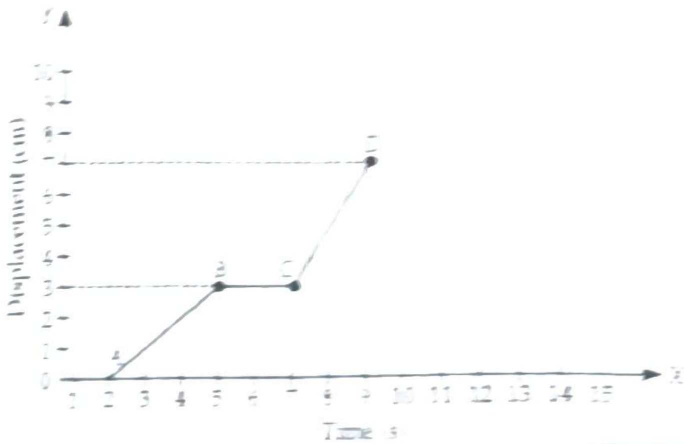
2. Distance-time graph for the motion of an object is shown in figure. Calculate the speed (in m s^{-1}) of the object at t = 2 second.



3. Find the speed (in m s^{-1}) of an object which has travelled 360 m in 1 minute.

Case Based Questions

Case I : A moving body changes its position continuously with time. Motion of a moving body can be easily described by drawing displacement-time graph. Difference between distance - time graph and displacement - time graph is that the slope of distance-time graph can be positive, negative or zero. Displacement-time of moving car is shown below.



CFQs

1. Find the speed of the body as it moves from A to B.

(a) 1 cm/s (b) 0.1 cm/s
(c) 1 m/s (d) 0.1 m/s

2. The slope from B to C is

(a) zero (b) 2 (c) $\frac{2}{3}$ (d) $\frac{3}{2}$

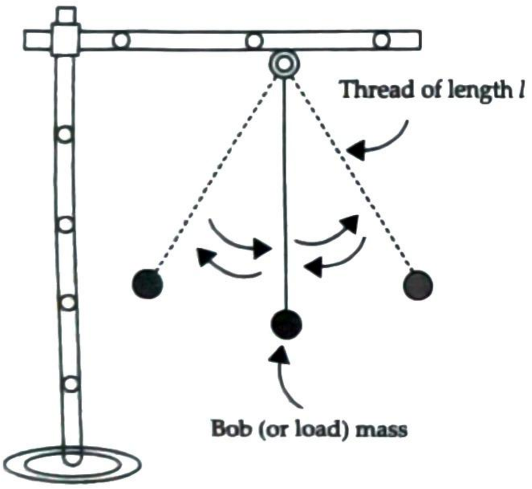
3. In case of moving body

(a) displacement = distance
(b) displacement ≥ distance
(c) displacement ≤ distance
(d) distance < displacement

4. The displacement-time graph of body is a straight line inclined to time axis. The body is in
   (a) uniform motion
   (b) uniformly accelerated motion
   (c) uniformly retarded motion
   (d) cannot predict.

5. The speed of the body as it moves from C to D, is
   (a) 2 cm/s (b) zero
   (c) 5 cm/s (d) 9 cm/s

Case II : An ideal simple pendulum consists of a point-mass suspended by a flexible, inelastic and weightless string from a rigid support of infinite mass. In practice, we can neither have a point-mass nor a weightless string.



In practice, a simple pendulum is obtained by suspending a small metal bob by a long and fine cotton thread from a rigid support. In the equilibrium position, the bob of a simple pendulum lies vertically below the point of suspension. If the bob is slightly displaced on either side and released, it begins to oscillate about the mean position.

For small oscillations, the motion of the bob is simple harmonic. Its time period is

$$T = 2\pi\sqrt{\frac{l}{g}}$$

The time period of a simple pendulum de
on its length l and acceleration due to gra
but is independent of the mass m of the

1. What will be the period of oscillation, length of a second's pendulum is ha
   (a) $\sqrt{2}$ s (b) 2 s
   (c) $\sqrt{3}$ s (d) $2\sqrt{2}$ s

2. What is the length of a simple pendu which ticks seconds?
   (a) 0.992 m (b) 0.992 cm
   (c) 1.22 cm (d) 1.22 m

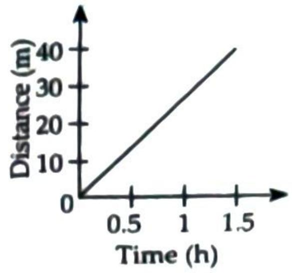
3. If the bob of a simple pendulum is ma wood, then the effect on the time per the wooden bob is replaced by an ide bob of iron
   (a) increases (b) does not ch
   (c) decreases
   (d) first increases then decreases.

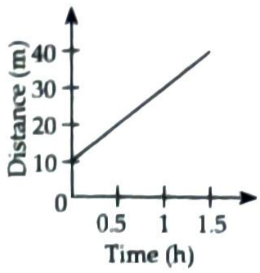
Case III : An object moving along a str line path is said to have uniform motion, speed remains constant. An object mo along a straight line path is said to have uniform motion, if its speed keeps chan Below table shows distance travelled by bodies in different time intervals.

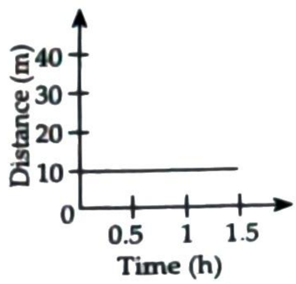
Time	Body A (Distance/m)	Body B (Distance
12.00 (noon)	10	12
12.15 pm	15	15
12.30 pm	20	19
12.45 pm	25	27
1.00 pm	30	39
1.15 pm	35	45
1.30 pm	40	48

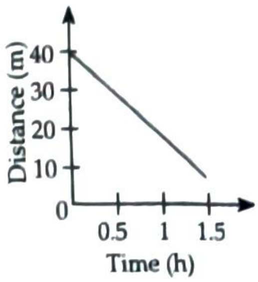
1. The motion of body A is
   (a) non-uniform
   (b) uniform
   (c) circular
   (d) uniform till 12 : 30 pm then non-unifo

2. Choose the correct distance time graph from the given table for body A.

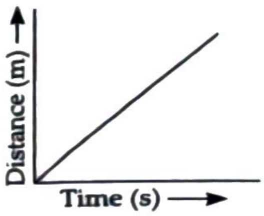
(a) 

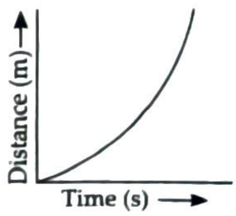
(b) 

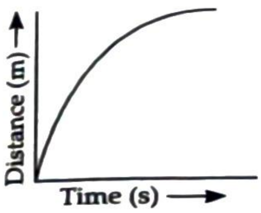
(c) 

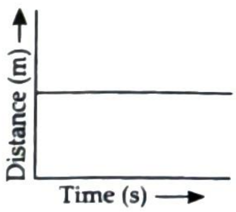
(d) 

3. Which of the following graph represent uniform motion of a moving object correctly?

(a) 

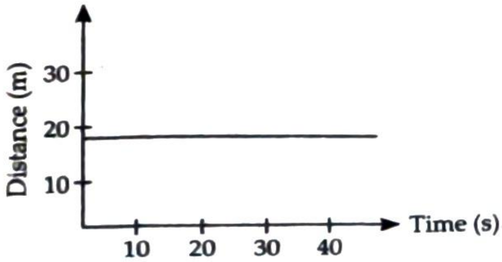
(b) 

(c) 

(d) 

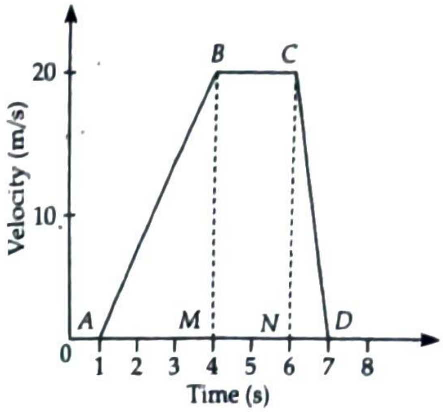
4. The motion of body B is
(a) uniform
(b) straight
(c) non-uniform
(d) periodic motion.

5. From the given graph, the body is



(a) at rest
(b) randomly moving
(c) in uniform motion
(d) in non-uniform motion.

Case IV : The variation of velocity of an object can be plotted against the time. Such a graph is known as velocity-time graph. The slope of the velocity-time graph gives the acceleration of the object and the area under the velocity-time curve gives the displacement of the object.



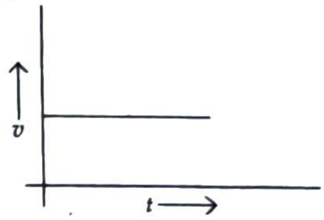
1. Find the acceleration between A and B
(a) 66.7 m/s^2
(b) 6.67 m/s
(c) 6.67 m/s^2
(d) zero

2. The average speed for the complete journey from 1 s to 7 s is
(a) 9.1 m/s
(b) 1.3 m/s
(c) $\frac{10}{3}$ m/s
(d) 13.33 m/s

3. The acceleration between B and C is
(a) 1 m/s^2
(b) 2 m/s^2
(c) 6.67 m/s^2
(d) zero

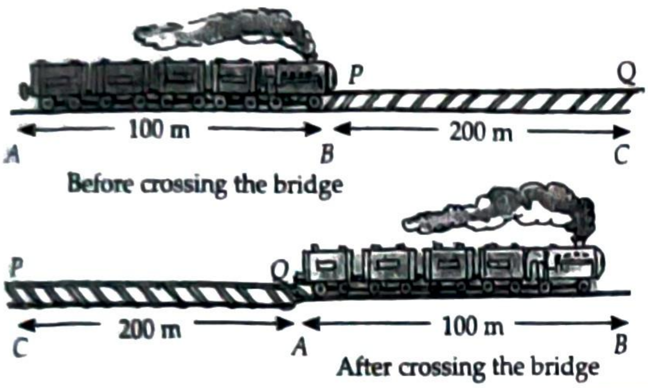
4. The velocity-time graph of a particle is not a straight line. Its acceleration is
(a) zero
(b) constant
(c) negative
(d) variable.

5. From the given $v-t$ graph, it can be inferred that the object is



(a) in uniform motion
(b) at rest
(c) in non-uniform motion
(d) moving with uniform acceleration.

Case V : A train crosses a bridge with constant velocity in 50 s as shown here. As the train crosses the bridge, the train covers a distance equal to its length and additional distance equal to the length of the bridge to cross it.



CFQs

1. The distance travelled by the train i
   (a) 30 m (b) 100 m
   (c) 300 m (d) 150 m
2. The velocity of the train is
   (a) 5.33 m/s (b) 9.5 m/s
   (c) 6 m/s (d) 2 m/s
3. Choose the correct statement:
   (a) The magnitudes of speed and veloc same when a body travels in a st line path.
   (b) Average speed of a moving body c equal to zero, but its average vel cannot be equal to zero.
   (c) To describe the velocity, directi necessary.
   (d) Both (a) and (c)
4. If the train is moving with velocity 54 k then the time taken to cross the bridg
   (a) 2 s (b) 5 s (c) 10 s (d) 2

Light : Shadows and Reflections

CHAPTER
04

Learning Outcomes...

- Reflection and Refraction
- Source of Light and Shadows
- Plane Mirror
- Spherical Mirror
- Lenses
- Pinhole Camera and Newton's Disc

Introduction

Light is a form of energy. Light is needed to see things around us. Light enables us to see objects from which it comes or from which it is reflected. We detect light with our eyes.

Sources of Light

Light sources are an essential part of our lives. They provide us with illumination, heat, and a variety of other benefits. The Sun is the main source of light and heat energy on the Earth. The sources of light can be classified into two types: **natural sources** of light and **artificial sources** of light.

Natural Light Sources: Natural sources of light produce light naturally and exist in nature on their own. They do not have human involvement in their production. The Sun is the main source of light on the Earth. The other natural sources are stars, lightning and fires from natural causes like the eruption of volcanoes.

Artificial Light Sources: Artificial sources of light do not produce light of their own. They emit light as a result of human activity. Electric bulbs, fluorescent bulbs, LEDs, fireworks, lasers, lighters, candles and torch are the examples of artificial source of light. Some common artificial light sources include:

- **Incandescent light bulbs**: These bulbs produce light by heating a filament until it glows. Incandescent light bulbs are relatively inefficient, meaning that they produce a lot of heat and waste a lot of energy.

to 57336.0 km? Calculate the speed of the car in km/min during this time. Express the speed in km/h also.

Long Answer Type

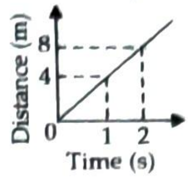
1. Explain sundial, water clock and sand clock with diagram.
2. What is simple pendulum? Explain its working.
3. Explain with the example, how can we choose a suitable scale.
4. There are bodies in motion which execute two or more types of motion at the same time. Give two examples and explain.
5. The following is the distance-time table of a moving car.

Time	Distance
10.05 am	0 km
10.25 am	5 km
10.40 am	12 km
10.50 am	22 km
11.00 am	26 km
11.10 am	28 km
11.25 am	38 km
11.40 am	42 km

(a) Use a graph paper to plot the distance travelled by the car versus the time.
(b) When was the car travelling at the greatest speed ?
(c) What is the average speed of the car ?
(d) What is the speed between 11.25 am and 11.40 am ?
(e) During a part of the journey, the car was forced to slow down to 12 km/h. At what distance did this happen ?

Integer/Numerical Value Type

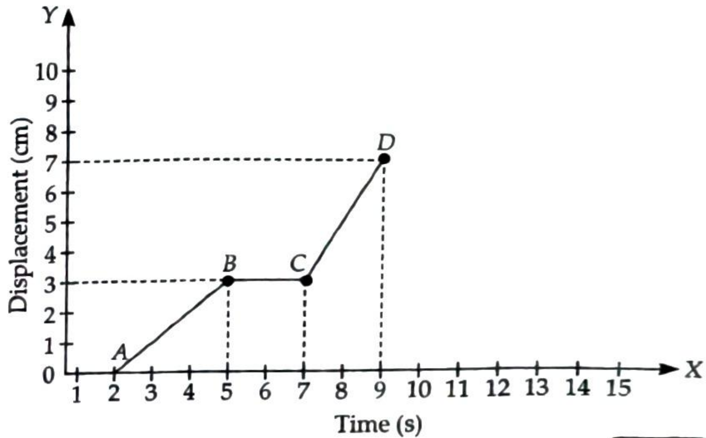
1. An electric train is moving with a velocity of 120 km/h. How much distance (in km) will it move in 30 s?
2. Distance-time graph for the motion of an object is shown in figure. Calculate the speed (in m s^{-1}) of the object at $t = 2$ second.



3. Find the speed (in m s^{-1}) of an object which has travelled 360 m in 1 minute.

Case Based Questions

Case I : A moving body changes its position continuously with time. Motion of a moving body can be easily described by drawing displacement-time graph. Difference between distance - time graph and displacement - time graph is that the slope of distance-time graph can be positive, negative or zero. Displacement-time of moving car is shown below.

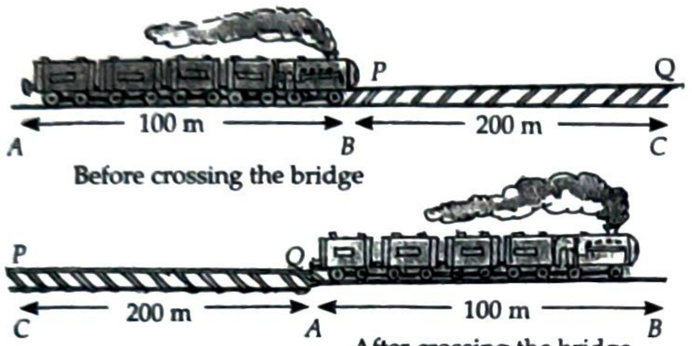


CFQs

1. Find the speed of the body as it moves from *A* to *B*.
   (a) 1 cm/s (b) 0.1 cm/s
   (c) 1 m/s (d) 0.1 m/s
2. The slope from *B* to *C* is
   (a) zero (b) 2 (c) $\frac{2}{3}$ (d) $\frac{3}{2}$
3. In case of moving body
   (a) displacement = distance
   (b) displacement ≥ distance
   (c) displacement ≤ distance
   (d) distance < displacement

(a) in uniform motion
(b) at rest
(c) in non-uniform motion
(d) moving with uniform acceleration.

Case V : A train crosses a bridge with constant velocity in 50 s as shown here. As the train crosses the bridge, the train covers a distance equal to its length and additional distance equal to the length of the bridge to cross it.



CFQs

1. The distance travelled by the train is
   (a) 30 m (b) 100 m
   (c) 300 m (d) 150 m
2. The velocity of the train is
   (a) 5.33 m/s (b) 9.5 m/s
   (c) 6 m/s (d) 2 m/s
3. Choose the correct statement:
   (a) The magnitudes of speed and velocity are same when a body travels in a straight line path.
   (b) Average speed of a moving body can be equal to zero, but its average velocity cannot be equal to zero.
   (c) To describe the velocity, direction is necessary.
   (d) Both (a) and (c)
4. If the train is moving with velocity 54 km/h, then the time taken to cross the bridge is
   (a) 2 s (b) 5 s (c) 10 s (d) 20 s

- Since, displacement is measured from the initial position to the final position, it has a direction. Therefore, it is a quantity that has both magnitude and direction.

Velocity

- When we talk about motion of a body, we should not only know about its speed but also the direction in which the body is moving.
- Velocity is the rate at which an object changes its position. It can be referred as speed in a particular direction. It is defined as the distance travelled by a body in unit time in a given direction.
- Velocity can also be defined as displacement per unit time.

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}}$$

- The SI unit of velocity is same as that of the speed *i.e.*, m s^{-1}.
- Speed is the distance travelled by an object in unit time and velocity is the distance travelled by an object in unit time in a given direction. Speed of a moving body can never be zero but velocity of a moving object will be zero, if it returns to its original position, *i.e.*, when its displacement is zero.

Acceleration

- In non-uniform motion, the speed and the velocity varies with time. The rate at which velocity varies is called acceleration. Thus, acceleration is the change in velocity per unit time.
- If the velocity of an object varies from an initial value u to the final value v in time t, then acceleration is given by $a = \frac{v-u}{t}$.
- The S.I. unit of acceleration is m s^{-2} or (m/s^2).
- The non-uniform motion is also called accelerated motion.

ILLUSTRATIONS

7 An express train leaves New Delhi at 6:00 pm and reaches Lucknow at midnight. Its speed is 80 km/h. What is the distance between Delhi and Lucknow?

Ans.: Speed = 80 km/h, time taken = 6 h.
In one hour it travels 80 km. Thus in six hours it travels 80 × 6 = 480 km. Thus, the distance between Delhi and Lucknow is 480 km.

8 Arrange the following speeds in an increasing order:

(a) A bicycle moving with a speed of 18 km/h,
(b) A fast runner running with a speed of 7 m/s,
(c) A car moving with a speed of 2000 m/min.

Ans.: Speeds of different moving bodies can be compared only when all these are expressed in the same unit. Let us express all the speeds in the SI unit, *i.e.*, in m s^{-1}.

(a) Speed of bicycle

$$= \frac{18 \text{ km}}{1\text{h}} = \frac{18 \times 1000 \text{m}}{1 \times 60 \times 60 \text{ s}} = \frac{18000 \text{ m}}{3600 \text{ s}} = 5 \text{ m s}^{-1}$$

(b) Speed of a runner = 7 m s^{-1}

(c) Speed of car

$$= \frac{2000 \text{m}}{1 \text{min}} = \frac{2000 \text{ m}}{1 \times 60 \text{ s}} = 33.3 \text{ m s}^{-1}$$

The speeds in the increasing order are 5 m s^{-1}, 7 m s^{-1} and 33.3 m s^{-1}.
This means, the speeds follow the order: Bicycle, Fast runner, Car *i.e.*, 18 km/h, 7 m/s, 2000 m/min (Increasing order)

9 A man riding a scooter travels a distance of 60 metres in 20 seconds. What is the speed of the scooter?

Ans.: $\text{Speed} = \frac{\text{Distance travelled}}{\text{Time taken}}$

Here, distance travelled = 60 m

Time taken = 20 s ; Speed $= \frac{60 \text{ m}}{20 \text{ s}} = 3 \text{ m s}^{-1}$

Thus, the speed of scooter is 3 metres per second.